Phase 2: Information and tasks

Status

It is now the 10th of January 2027 and the measles outbreak has continued to grow in all countries in Scandiland and abroad. It is unclear how the Christmas holiday has affected spread, but it is now clear that the outbreak is a significant health crisis and it has been declared a Public Health Emergency of International Concern (PHEIC) by the WHO.

The public health authorities urgently need advice about the further development of the outbreak and needs to assess the effect of possible interventions. Data sharing between Scandiland countries has continued and an updated line-list of all cases has been provided. By now there are a significant number of cases in each country.

Country	Reported Cases
Denden	53,252
Sweway	4,867
Finmark	2,252
Norland	1,640

Tomorrow, Scandiland governments will meet to discuss the current situation and make decisions about possible interventions. The modelling teams will need to provide input to help them make the best decisions possible. Interventions that are currently under consideration includes:

- School closures - Close all schools for the duration of the outbreak
- Work from home - Mandate all non-critical employees to work from home for the duration of the outbreak. It is expected that 30% can work from home.
- Re-vaccination with the existing measles vaccine - Recent laboratory research indicates that re-vaccination with the current vaccine can give some protection against infection with a VE of approximately 20%. Each country in Scandiland have access to enough vaccines to vaccinate approximately 15% of their population.
- Do nothing - Let the outbreak run it's course without government mandated interventions.
- Other suggestions from the modelling teams

Authorities want to understand how each of these options will impact the overall disease burden of the outbreak, with a focus on hospital capacity. While hospital capacity is difficult to measure, modelling teams have been advised that if the number of people hospitalized concurrently exceeds more than 80 per 100,000 the health system will struggle significantly.

Information

- An updated line-list as of the 10th of January is now available.
- A seroprevalence survey has been conducted in the Vejager area of Denden in December. Analysis of the data provides an estimate of the fraction of the population that had been infected prior to this time. More information about the survey is available together with the data in the repository.
- A contact survey has been performed for the Scandiland-region. Contacts broken down by age were similar across the region and a contact matrix has been estimated and made available.

All data for this exercise can be found at:
https://github.com/Gulfa/modelling_prep_ex/tree/main/exercise_data

Tasks - update

Public health authorities requests modelling results to support intervention decisions in each country. They need an assessment of the likely and reasonable worst case scenario for the future of the outbreak without any interventions and assessments of the effects of the interventions above. It is crucial to assess the uncertainty in the scenarios and to provide understandable communications to decision makers.

It will be up to the modelling teams to prioritize their work to provide the best possible advice in the available time. It may not be feasible to consider all the interventions mentioned above.

The modelling teams have until 12:00 to prepare a presentation of their assessment of the current situation and scenarios with and without interventions. **Each group will get 8 minutes to present**.